Cyclone Dossier

Historical Ground-Truth Reference: Cyclone Biparjoy (2023) & Cyclone Amphan (2020)

Prepared for: ML cyclone-tracking / pattern-classification project — structured-data track (IBTrACS best-track ground truth, event background, MoES/IMD context)

Scope: North Indian Ocean basin, Regional Specialized Meteorological Centre (RSMC) New Delhi (India Meteorological Department) area of responsibility.

Compiled from: Wikipedia (cyclone articles), NASA Earth Observatory, IMD/JTWC post-storm reporting summaries, peer-reviewed post-storm analyses (arXiv, Springer), and contemporary news coverage. All figures should be cross-checked against the official IBTrACS CSV / IMD best-track reports before use in model validation.

1. Extremely Severe Cyclonic Storm Biparjoy (2023)

Biparjoy was a long-lived, erratic tropical cyclone over the Arabian Sea — the third depression and second cyclonic storm of the 2023 North Indian Ocean cyclone season. Its name means “disaster”/“calamity” (Bengali). It is notable for an unusually long duration over open water (about 13 days) and multiple intensification/weakening cycles before landfall in Gujarat.

1.1 Key Statistics

Field	Value
Formed	6 June 2023 (depression, IMD)
Dissipated	19 June 2023 (weakened to well-marked low)
Peak classification (IMD)	Extremely Severe Cyclonic Storm
Peak classification (JTWC)	Category 3-equivalent tropical cyclone
Highest winds — IMD (3-min sustained)	165 km/h (105 mph)
Highest winds — JTWC (1-min sustained)	205 km/h (125 mph)
Lowest central pressure (IMD)	958 hPa
Lowest central pressure (JTWC)	944 hPa
Landfall location	Near Jakhau Port / Naliya, Gujarat, India (Mandvi–Karachi stretch)
Landfall date/time	15–16 June 2023 (≈19:00–22:00 IST, 15 June)
Landfall intensity	Very Severe Cyclonic Storm; sustained ~115–140 km/h, gusts to ~140–145 km/h
Basin / Sub-basin	North Indian Ocean — Arabian Sea (ARB)
Areas affected	India (Gujarat), Pakistan (Sindh)
Fatalities	17 total
Injuries	24
Estimated damage	US$148 million (2023 USD)
People evacuated	>170,000–180,000 across India & Pakistan (~94,000+ in Gujarat alone)

1.2 Meteorological Timeline

Date	Stage	Basin position	Max winds (approx.)	Notes
6 Jun 2023	Depression	East-central Arabian Sea	—	First noted by IMD
8–9 Jun 2023	Cyclonic Storm	Arabian Sea, moving N/NNW	65–85 km/h	Slow northward track begins
11–12 Jun 2023	Very Severe Cyclonic Storm	East-central Arabian Sea	~150–165 km/h	Rapid strengthening phase; IMD issues evacuation alerts for Gujarat
14 Jun 2023	Category 1 (SSHWS)	Approaching Gujarat/Sindh coast	~129 km/h	NOAA-20/VIIRS imagery captured; slow ~8 days spent over open Arabian Sea
15 Jun 2023 (evening)	Landfall	Near Jakhau Port, Gujarat	115–140 km/h, gusts ~140–145 km/h	Landfall process began evening, continued to ~midnight
16 Jun 2023	Weakening / Depression	Interior Gujarat/Rajasthan	~78 km/h and falling	Wind speed dropped sharply post-landfall
19 Jun 2023	Dissipated	Well-marked low-pressure area	—	System fully dissipated

Structural note for pattern-classification labeling: Biparjoy is a good “slow evolution / long track” test case — it shows an eye/eyewall-dominant structure at peak intensity (11–13 June), followed by a shear-affected, asymmetric phase as it accelerated northeast and interacted with drier air near the coast, then a rapid landfall-degrading collapse after 15 June.

1.3 Impact Summary

- Roofs blown off, thousands of trees and ~700+ electricity poles uprooted across eight affected Gujarat districts (Kutch, Saurashtra region).
- Storm surge and heavy rainfall caused coastal flooding; rail network disruptions reported.
- No large-scale loss of life at landfall itself (attributed to large-scale pre-emptive evacuation); most fatalities were indirect (drowning, wall collapse, accidents).
- Both India and Pakistan carried out coordinated mass evacuations before landfall — relevant context for any “warning lead-time” discussion vs. IMD's operational forecasts.

2. Super Cyclonic Storm Amphan (2020)

Amphan was the first super cyclonic storm to form in the Bay of Bengal since the 1999 Odisha cyclone, and the first named tropical cyclone of the 2020 North Indian Ocean season. It is the defining “rapid intensification” case in the region — strengthening from a cyclonic storm to a super cyclone in under 36 hours — making it a strong contrast case against Biparjoy's slow, long-duration evolution.

2.1 Key Statistics

Field	Value
Formed	16 May 2020 (depression, IMD) / low-pressure area noted 13 May
Dissipated	21 May 2020
Peak classification (IMD)	Super Cyclonic Storm
Peak classification (JTWC)	Category 5-equivalent tropical cyclone
Highest winds — IMD (3-min sustained)	240 km/h (150 mph)
Highest winds — JTWC (1-min sustained)	270 km/h (165 mph)
Lowest central pressure (IMD)	920 hPa
Lowest central pressure (JTWC)	901 hPa
Landfall location	Between Digha (West Bengal, India) and Hatiya (near Sundarbans) / India-Bangladesh border region
Landfall date/time	20 May 2020, ≈10:00–11:00 UTC (afternoon IST)
Landfall intensity	Very Severe Cyclonic Storm; sustained ~155–165 km/h, gusts to ~185–200 km/h
Basin / Sub-basin	North Indian Ocean — Bay of Bengal (BOB)
Areas affected	India (West Bengal, Odisha, Andaman Is.), Bangladesh, Sri Lanka, Bhutan, Myanmar, Thailand
Fatalities	133 total
Estimated damage	US$15.5 billion (2020 USD) — 2nd costliest cyclone on record in North Indian Ocean
Track length	~1,820 km total (~1,328 km over ocean, ~492 km over land)

Field	Value
People affected	~18 million; ~3 million houses damaged; ~449,000 electric poles downed

2.2 Meteorological Timeline

Date	Stage	Basin position	Max winds (approx.)	Notes
13 May 2020	Low-pressure area	~300 km east of Colombo, Sri Lanka	—	Genesis disturbance over exceptionally warm SST
15–16 May 2020	Tropical Depression → Depression	~11.4°N, 86.0°E, Bay of Bengal	<60 km/h	JTWC upgrade 15 May; IMD follows 16 May
17 May 2020	Severe Cyclonic Storm	Central Bay of Bengal	80–120 km/h, gusts 140 km/h	Rapid intensification begins (within 12 hrs → extremely severe)
18 May 2020, ~12:00 UTC	Peak intensity — Super Cyclonic Storm	Central Bay of Bengal	240 km/h (IMD 3-min) / 270 km/h (JTWC 1-min)	Category 5-equivalent; only super cyclone in region in two decades
19 May 2020	Weakening, still Category 3-equiv.	Approaching West Bengal/Bangladesh coast	~185 km/h	Moving north-northeast over Bay of Bengal
20 May 2020, 10–11 UTC	Landfall	Between Digha (WB) and Sundarbans/Hatiya	155–165 km/h, gusts 185–200 km/h	Landfall as Very Severe Cyclonic Storm; passed close to Kolkata
21 May 2020	Dissipated	Interior Bangladesh/ Northeast India	—	Rapid weakening over land

Structural note for pattern-classification labeling: Amphan is the key “rapid intensification / eye ” test case — well-defined eye/eyewall structure visible at and near peak intensity (17–19 May), useful for training the 'Eye / Eyewall' and 'Banding' classes; the 19–20 May approach-to-landfall segment shows gradual weakening/asymmetry as it interacted with the coastline, transitioning into the 'Landfall-degrading' class.

2.3 Impact Summary

- Second-costliest cyclone on record in the North Indian Ocean (India alone reported ~US$13.2 billion in damages in West Bengal).

- ~3 million houses damaged; ~18,000 sq km of agricultural land and mangrove forest affected.
- Response was significantly complicated by the concurrent COVID-19 pandemic (shelter capacity limits, mobility restrictions, PPE requirements) — relevant context if discussing operational/response gaps rather than pure forecast-accuracy gaps.
- Widespread storm-surge flooding and destructive winds extended impact across India, Bangladesh, Sri Lanka, Bhutan, Myanmar, and Thailand.

3. Side-by-Side Comparison

Attribute	Biparjoy (2023)	Amphan (2020)
Basin	Arabian Sea	Bay of Bengal
Duration	~13 days (long-lived)	~5–6 days (fast-evolving)
Peak IMD category	Extremely Severe Cyclonic Storm	Super Cyclonic Storm
Peak 3-min wind (IMD)	165 km/h	240 km/h
Lowest pressure (IMD)	958 hPa	920 hPa
Intensification pattern	Gradual, multiple fluctuation cycles	Rapid intensification (<36 hrs to super cyclone)
Landfall state	Gujarat, India (west coast)	West Bengal, India (east coast, near Bangladesh)
Fatalities	17	133
Damage (USD)	~$148 million	~$15.5 billion
Best-track ML use case	Slow-evolution, long-track / erratic-motion stress test	Rapid-intensification / peak-eye-structure stress test

4. Notes for Structured-Data / Ground-Truth Work

- **IBTrACS access:** Pull `ibtracs.NI.list.v04r00.csv` (North Indian basin subset) from NOAA NCEI. Filter by `NAME` = BIPARJOY / AMPHAN and `SEASON` = 2023 / 2020 respectively. Row 2 of the raw file is a units row — skip it when loading with pandas.
- **IMD-specific columns:** Use the `NEWDELHI_LAT` / `NEWDELHI_LON` / `NEWDELHI_WIND` / `NEWDELHI_PRES` fields for an apples-to-apples “model vs. IMD” comparison, since these are IMD's own reported positions/intensities (RSMC New Delhi is the official reporting agency for this basin) rather than a multi-agency blended value.
- **Timestamps:** Best-track points are generally at 3-hour intervals for RSMC New Delhi reports (6-hourly for many other agencies) — confirm cadence before computing MAE so predicted and ground-truth timestamps are properly aligned/interpolated.
- **Caveat:** All figures in this dossier are compiled from secondary sources (Wikipedia, NASA, news, peer-reviewed post-storm papers) for quick reference and framing. Before final MAE/validation work or slide claims, cross-check exact per-timestamp positions and intensities directly against the downloaded IBTrACS CSV and, where possible, official IMD cyclone e-Atlas / RSMC best-track reports — those are the authoritative sources of record.

Sources: Wikipedia — “Cyclone Biparjoy”, “Cyclone Amphan”; NASA Earth Observatory (VIIRS/MODIS imagery captions); IMD/JTWC operational bulletins as summarized in contemporary reporting (Al Jazeera, CBS News, Deccan Herald, Gulf News); peer-reviewed post-storm analyses (arXiv:2310.18114, arXiv:2007.02982, arXiv:2012.04384, Springer s41976-021-00048-z). Compiled 28 August 2026.